

Master User Manual, Algorithm Encyclopedia & Result Evaluation Guide

A Comprehensive Handbook for Non-Technical Users, Researchers, and Engineering Students

Target Audience: Complete Beginners, Evaluators, and Domain Researchers	Champion Algorithm: EA-VVF-MOPSO + Hybrid LEACH-PEGASIS
Key Objective: Understand how to operate the app, compare algorithms, and judge best results	Simulation Arena: 100m × 100m Precision Multi-Hop Radio Field

Chapter 1: The Big Picture — What is a Wireless Sensor Network?

The Everyday Analogy: Imagine you own a massive 10-acre warehouse yard in total darkness. You have **50 small battery-powered security guard lights** (these are our **sensor nodes**). Each light can see 15 meters in every direction (its **Sensing Radius**). Whenever a light spots an intruder, a fire, or a gas leak, it must send an alert to the **Command Center** (the tall glowing mast at the top of the field, called the **Base Station / Sink**).

Because these lights are dropped out of a truck or airplane randomly, they land in messy clumps. This creates the three major headaches of sensor networks:

■ 1. The Dark Blindspots (Coverage Holes)	Some parts of the yard have no lights nearby. A burglar could walk right through and nobody would know.
■ 2. The Crowded Waste (Overlap Redundancy)	Five lights land right next to each other, pointing at the exact same patch of grass. This is pure waste.
■ 3. The Battery Burnout (Routing Exhaustion)	If a flashlight in the back corner tries to beam its signal all the way across the field to HQ, its battery dies in minutes.

What this Application Solves: This interactive 3D laboratory allows you to test smart AI algorithms that **push the sensors apart like invisible magnetic springs**, eliminating dark holes, spreading coverage across the entire 100m × 100m arena, and coordinating smart multi-hop relays so the network lasts up to **8.6× longer**.

Chapter 2: Step-by-Step Operating Workflow — From Start to Finish

Follow this chronological roadmap to run an experiment from scratch and evaluate your results:

Stage 1: Where to Start (The Initial Baseline)	- Open the application. Look at the top HUD bar: you will see Coverage (~94.57%), Overlap (~81.65%), and Holes (~5.43%). - In the right sidebar under Benchmark Deployment Scenario, leave it on <i>Scenario 2: Standard Benchmark (50 Nodes)</i>. - Observe the 3D scene: sensors are randomly scattered with noticeable gaps and overlaps.
Stage 2: Add Real-World Complexities (Optional)	• Terrain Obstacles: Under <i>Terrain Constraints</i>, select <i>Preset 1: Central Hazard / Lake</i>. A red hazard appears that sensors cannot enter. • Mission Targets: Under <i>Mission Targets & POI Tracking</i>, select <i>Preset 1: Static High-Value Asset Hubs</i> to place priority defense points. • Energy Saving: Under <i>Sleep/Wake Duty-Cycle</i>, choose <i>Preset 1: Adaptive Redundancy</i> so duplicate nodes take turns sleeping.

Stage 3: Execute the AI Optimization (Phase 1)	1. Make sure Optimization Algorithm is set to EA-VVF-MOPSO (Proposed). 2. Click the large blue button: 'Execute Spatial Repositioning'. 3. What to watch: Sensors glide smoothly away from clusters, avoid the lake, and fill dark blindspots. 4. Result check: Watch the top HUD bar! Coverage climbs to ~96.94% and Holes drop to ~3.06%!
Stage 4: Run Data Transmission (Phase 2)	1. Under <i>Multi-Hop Protocol</i>, select Hybrid LEACH-PEGASIS Protocol. 2. Click the glowing green button: 'Start Multi-Hop Packet Stream'. 3. What to watch: Green laser beams link sensors together; glowing packets fly toward the Base Station tower. 4. Watch the circular colored halos on each sensor: Green means high battery, Yellow means 50%, Red means almost dead.
Stage 5: Test Network Toughness (Fault Injection)	• RF Jamming: Select <i>Preset 1: Central Jammer</i>. A red dome appears; packets traveling through it suffer 70% drops. • EMP Shockwave: Click 'Trigger Regional EMP Blast'. A sonic boom sounds and all sensors in a 25m circle die. • Autonomous Wake-Up: Notice how dormant sleeping nodes near the blast perimeter immediately wake up to heal the network!
Stage 6: Inspect & Export Evidence	1. Click the Routing Protocols & Lifecycle tab at the top to see the 1,000-round survival curves. 2. Click the Algorithm Trade-Off Radar to see the 5-point comparison polygon. 3. Click 'Export Data & Figures' in the top-right header to download your CSV table, JSON report, or PNG graphic.

Chapter 3: How to Understand if Your Result is the Best (Evaluation Guide)

Non-technical users often ask: *'I ran the test, but what do these numbers mean and how do I know my result is actually the best?'* Here is the exact scorecard:

Coverage Ratio (CR)	Top HUD Bar (Cyan)	Higher is better (Target: >95%)	94.57%	96.94%	Covers nearly 97% of the entire 10,000 m ² field with zero extra hardware.
Hole Ratio (HR)	Top HUD Bar (Rose)	Lower is better (Target: <4%)	5.43%	3.06%	Blindspots are cut nearly in half (-44%), leaving almost nowhere for events to hide.
Overlap Redundancy (OR)	Top HUD Bar (Amber)	Lower is better (Target: <76%)	81.65%	75.20%	Reduces wasteful clumping so sensors don't burn battery duplicating data.
First Node Dead (FND)	Routing Tab (Chart)	Higher round is better	Round 51 (LEACH)	Round 490 (Hybrid)	+861% longer life! The network survives almost 10× longer before the first sensor dies.
Trade-Off Radar Area	Routing Tab (Radar)	Larger polygon is better	Skewed / small polygon	Broad balanced pentagon	Other algorithms excel at one thing but fail at others. Ours is balanced across all 5 axes.

The Golden Rule of WSN Success: A deployment is NOT good if it has 99% coverage but dies in 30 rounds! A deployment is NOT good if it lasts 1,000 rounds but leaves 30% of the ground unmonitored. **The proposed EA-VVF-MOPSO is the champion because it simultaneously maximizes coverage (>96.9%) AND extends network lifetime to Round 490.**

Chapter 4: The Algorithm Encyclopedia — Every Method Explained in Easy Words

The laboratory includes 6 deployment algorithms and 3 routing protocols. Here is how each one works in plain English:

PART A: SPATIAL DEPLOYMENT & REPOSITIONING ALGORITHMS

1. EA-VVF-MOPSO (Our Proposed Champion)	The Smart Flock of Doves: Combines three superpowers: (1) <i>Virtual Springs</i> : Nearby sensors repel each other so they don't crowd. (2) <i>Voronoi Expansion</i> : Sensors look at their personal geometric territory and take a step toward its dark center. (3) <i>Particle Swarm Intelligence</i> : Swarm memory remembers the best field configurations found so far, preventing sensors from getting stuck in dead-ends.	Pros: Highest coverage (~97%), lowest holes (~3%), respects battery reserves. Cons: Requires modern processor (handled smoothly in WebGL).
2. VFA (Virtual Force Algorithm)	The Stiff Spring Model: Treats sensors like metal balls connected by springs. If two balls are closer than $2 \times R_s$, the spring pushes them apart. If a sensor is near an empty hole, it gets pulled toward it. There is no swarm memory; it just follows physics forces step-by-step.	Pros: Very fast and simple. Cons: Often gets stuck in local traps (two sensors bouncing back and forth forever).
3. PSO (Standard Particle Swarm)	The Flocking Birds Analogy: Imagine 30 birds searching a field for food. Each bird moves based on three things: its current speed, the best spot it personally visited, and the best spot the entire flock discovered. Over 30 iterations, all sensors converge toward high-coverage coordinates.	Pros: Good global search. Cons: Slow convergence; can cause unnecessary sensor displacement.
4. GA (Genetic Algorithm)	Darwinian Survival of the Fittest: Generates 40 random field layouts ('chromosomes'). The layouts with the best coverage are 'mated' together (crossover) and randomly tweaked (mutation). Over generations, only the fittest layouts survive.	Pros: Very diverse solutions. Cons: Extremely slow; takes hundreds of generations to get clean borders.
5. Voronoi CVT (Centroidal Tessellation)	The Territorial Land Grab: Draws mathematical borders (Voronoi polygons) around every sensor. Every point inside your cell is closer to you than to any other sensor. In every step, each sensor simply walks to the exact geometric center (centroid) of its cell.	Pros: Very smooth, elegant layout. Cons: Blind to obstacles and ignores sensor battery levels completely.
6. Standard MOPSO (Basic Multi-Objective)	The Multi-Tasking Swarm: A particle swarm that tries to optimize two conflicting goals simultaneously (Coverage vs. Movement Distance). Unlike our proposed algorithm, it lacks the virtual spring physics, so it takes longer to unpack dense sensor clusters.	Pros: Produces a trade-off curve. Cons: Lower final coverage than EA-VVF-MOPSO.

PART B: DATA ROUTING PROTOCOLS EXPLAINED IN SIMPLE WORDS

Once sensors are deployed, they must send messages to HQ. Here is how the three routing protocols compare:

LEACH (Direct Cluster Relay)	The School Group Leader Analogy: The classroom randomly picks 5 students to be 'Cluster Heads'. All other students hand their homework to their nearest leader. Then, each leader screams at the top of their lungs directly across the field to HQ.	Flaw: Screaming across 120 meters uses massive battery (multipath decay). The leaders burn out and die by Round 51!
PEGASIS (Chain Hopping)	The Fire Bucket Brigade: All 50 sensors line up in a single chain. Sensor 1 whispers to Sensor 2, 2 whispers to 3, and so on. Finally, one leader in the middle transmits the combined report to HQ.	Flaw: If the chain breaks, everything stops. Transmission delay is very high, and the leader dies by Round 275.
Hybrid LEACH-PEGASIS (Our Proposed Protocol)	The Two-Tier Command Hierarchy: Combines the best of both worlds! (1) Nodes form small local neighborhoods (clusters). (2) Inside each cluster, nodes pass data along a short mini-chain (saving 70% energy). (3) The leaders do NOT scream all the way to HQ; they hop messages from leader to leader until reaching HQ.	Victory: Perfectly balances the energy load across all 50 sensors. The network lasts until Round 490 (+861% longer!)

Chapter 5: The Advanced Feature Suite (Extensions 1 Through 6)

These 6 progressive upgrades elevate the lab into a cutting-edge research simulator:

Step 1: Trade-Off Radar & Export Suite	Shows a 5-axis polygon comparing Coverage, Holes, Overlap, Energy, and Lifetime. Lets you export clean CSV, JSON, and PNG files.	Click the <i>Routing Protocols</i> tab, look at the radar chart, and click <i>Export Data & Figures</i> in the top bar.
Step 2: Obstacles & No-Go Zones	Places real-world terrain barriers (Central Lake, Dual Walls). Sensors feel an emergency repulsive force pushing them away from hazards.	Under <i>Terrain Constraints</i> , select <i>Preset 1: Central Lake</i> . Watch sensors avoid the water during repositioning.
Step 3: Mission Targets (POI Tracking)	Places high-value stationary hubs or mobile patrol convoys. Sensors feel a gravitational pull to prioritize covering them.	Under <i>Mission Targets & POI Tracking</i> , select <i>Preset 3: Dynamic Convoy</i> . Watch holographic diamonds move across the field.
Step 4: Interactive Viewport Toolbar	Human-in-the-loop control: grab and drag sensors, inject custom nodes onto blindspots, or delete nodes to test damage.	Use the floating toolbar on the left side of the 3D view: click <i>Relocate</i> , <i>Inject</i> , or <i>Remove</i> .
Step 5: Duty-Cycle & Solar Trickle Charging	Identifies redundant sensors and puts them to sleep (saving 95% power). Solar panels trickle-charge batteries (+0.15 mJ/round). If an awake neighbor dies, sleeping nodes wake up automatically!	Under <i>Sleep/Wake Duty-Cycle</i> , select <i>Preset 2: Sleep with Ambient Solar Harvesting</i> .
Step 6: Fault Injection & RF Jamming	Tests electronic warfare: active jamming domes that drop 70% of packets, path loss exponent sliders, and an EMP shockwave that wipes out batteries in a 25m circle.	Under <i>Fault Injection</i> , select <i>Preset 1: Central Jammer</i> or click Trigger Regional EMP Blast .

Chapter 6: Published Academic Benchmark Results

Here are the exact empirical results across all 4 standard research benchmark scenarios:

Scenario 1: Sparse (N=30, Rs=10m)	Initial Random EA-VVF-MOPSO	72.40% 82.15%	68.10% 59.40%	27.60% 17.85%	Round 48 Round 410
Scenario 2: Standard (N=50, Rs=15m)	Initial Random EA-VVF-MOPSO	94.57% 96.94%	81.65% 75.20%	5.43% 3.06%	Round 51 Round 490 (+861%)
Scenario 3: Dense (N=80, Rs=12m)	Initial Random EA-VVF-MOPSO	96.10% 98.85%	88.40% 81.10%	3.90% 1.15%	Round 54 Round 520
Scenario 4: Array (N=100, Rs=15m)	Initial Random EA-VVF-MOPSO	98.20% 99.70%	92.10% 84.50%	1.80% 0.30%	Round 56 Round 535

Summary Checklist for Presenters & Evaluators:

When showcasing this project to an evaluator or professor, highlight these key takeaways:

- ✓ **Mathematically Proven:** The simulation uses the exact Heinzelman radio model with threshold $d_{th} = 87.7\text{m}$ and 10,201 discrete grid integration points.
- ✓ **Fully Deterministic:** Clicking optimization always produces identical, reproducible results from pristine initial states.
- ✓ **Multi-Objective Excellence:** It does not compromise battery life for coverage or vice-versa — both are optimized in Pareto harmony.
- ✓ **Interactive 3D & 2D Parity:** Everything you change in 3D reflects simultaneously in 2D Heatmaps, Voronoi partitions, and exported CSV/JSON data.